

MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
UNIVERSITY OF CAMBRIDGE LOCAL EXAMINATIONS SYNDICATE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME



CENTRE
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CHEMISTRY

9729/02

Paper 2 Structured Questions

October/November 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **19** printed pages and **1** blank page.



Singapore Examinations and Assessment Board



CAMBRIDGE
International Examinations



- 1 Phosgene, $\text{Cl}_2\text{C}=\text{O}$, is a colourless, toxic gas. It is used in the production of pesticides, rubbers and adhesives.

(a) (i) Complete Table 1.1 to describe one similarity and one difference between

- a 1s and a 2s orbital,
- a 2s and a 2p orbital.

Table 1.1

	similarity	difference
1s and 2s orbital	<u>Shape</u> of the orbitals. Both 1s and 2s orbitals are spherical in shape.	<u>Size</u> of the orbitals. The 2s orbital is larger than the 1s orbitals.
2s and 2p orbital	Both the 2s and 2p orbital can hold a <u>maximum of 2 electrons</u> each.	<u>Shape</u> of the orbitals: The 2s orbital is spherical while the 2p orbital is dumb-bell shape.

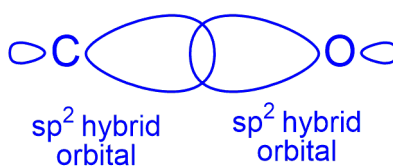
[2]

- (ii) State and explain, with reference to the Valence Shell Electron Pair Repulsion theory, the shape of a phosgene molecule.

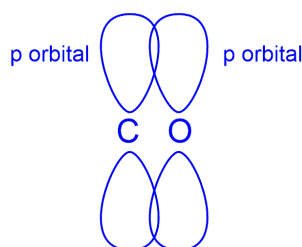
VSEPR theory states that the shape of a molecule is determined by the number of σ bond pair and lone pair of electrons around the central atom, which arrange themselves to be as far away as possible to minimise repulsion. There are 3 σ bond and 1 π bond pairs around the central C in phosgene, hence trigonal planar. [2]

- (iii) The molecule of phosgene contains both σ (sigma) and π (pi) bonds. Draw labelled diagrams to show how orbitals overlap to form

- a σ (sigma) bond



- a π (pi) bond.



[2]



(b) Table 1.2 shows the electronegativity values of the atoms in phosgene.

Table 1.2

atom	electronegativity / Pauling units
carbon	2.5
chlorine	3.0
oxygen	3.5

(i) Explain what is meant by the term *electronegativity*.

Electronegativity refers to the tendency of an atom to attract a shared (bonding) pair of electrons (or electron density) towards itself.

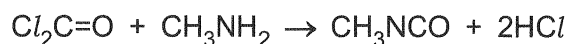
[2]

(ii) Predict all possible intermolecular forces which could exist between phosgene molecules. Explain how these forces arise.

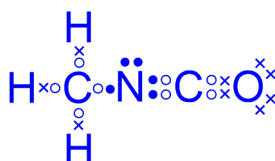
Due to movement of the electrons, the electron cloud of phosgene will be asymmetric at most instances, giving rise to an instantaneous dipole, which will induce a dipole in neighbouring molecules, resulting in attractive dispersion forces.

Due to electronegativity differences between C and O, and C and Cl, phosgene is a polar molecule with a permanent dipole moment. Opposite ends of the dipole in two neighbouring molecules will attract each other (permanent dipole-permanent dipole). [3]

(c) Phosgene reacts with methylamine to form methyl isocyanate.



(i) Draw a dot-and-cross diagram to show the bonding present within a methyl isocyanate molecule.

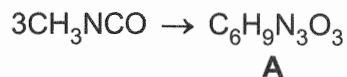


[1]

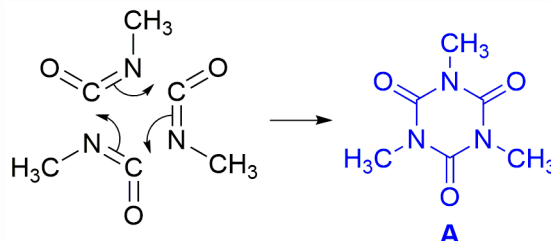




In the presence of a catalyst, methyl isocyanate reacts with itself to form compound **A**.



(ii) Molecules of **A** have no overall dipole moment. Suggest a possible structure of **A**.



[1]

(iii) Suggest why, at room temperature, methyl isocyanate is a gas but **A** is a solid.

As **A** is made up of 3 molecules of methyl isocyanate, **A** has three times the number of electrons and hence polarisability of the electron cloud is much higher, leading to much stronger instantaneous dipole-induced dipole attraction / dispersion forces between **A** and hence higher melting point.

[Total: 14]



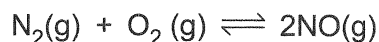


Question 2 starts on the next page.





- 2 A mixture of nitrogen, oxygen and nitrogen monoxide are placed in a 1.0 dm^3 sealed container and allowed to reach equilibrium.



The value of the equilibrium constant, K_p , measured at different temperatures is shown in Table 2.1.

Table 2.1

temperature/K	K_p
298	1×10^{-30}
700	5×10^{-13}
1100	4×10^{-8}
1500	1×10^{-5}

- (a) Explain what is meant by the term *dynamic equilibrium*.

Dynamic equilibrium refers to the state of a **reversible system** wherein the **rate of the forward and backward reactions are equal and constant**. [1]

- (b) Write the expression for the equilibrium constant, K_p , for this reaction.

$$K_p = \frac{p_{\text{NO}}^2}{p_{\text{N}_2} \cdot p_{\text{O}_2}}$$

[1]

- (c) Deduce the sign for the enthalpy change of the forward reaction. Explain your reasoning.

When **temperature is increased**, the system attempts to lower temperature by **favouring the endothermic reaction to absorb heat**. Since K_p increases with temperature, the forward reaction must be **endothermic**, hence enthalpy change has a **positive sign**. [1]



(d) At a constant temperature, the volume of the container is increased to 5.0dm^3 and the system allowed to reach equilibrium.

(i) Explain the effect this will have on the partial pressures of the individual gases and the composition of the reaction mixture.

As the system consists of equal number of gaseous particles on both sides of the equilibrium, the position of equilibrium is not affected by changes in total pressure, and so the **composition of the reaction mixture remains the same**. From the ideal gas equation, $pV = nRT$, when volume of the container is increased at constant temperature, **total pressure of the system will decrease**. Although the mole fraction of O_2 , N_2 and NO does not change, the **partial pressures of the individual gases will decrease**. [4]

(ii) Hence predict how the value of K_p will change.

The value of K_p **will not change**. [1]

(e) The value of K_p does **not** change when a catalyst is added. Explain why this is the case.

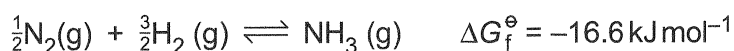
This is because a catalyst **increases the rate of the forward and backward reactions equally**, hence there will be **no change to the composition of the reaction mixture**, thus the value of K_p does not change. [2]

(f) Standard Gibbs free energy of formation, $\Delta G_f^\ominus$, refers to the free energy that accompanies the formation of one mole of substance in its standard state from its elements in their standard states (1 bar and 298K).

(i) Use the K_p value at 298K to suggest the sign and magnitude for $\Delta G_f^\ominus$ for the formation of nitrogen monoxide from its elements at 298K.

Since value of K_p is very small (1×10^{-30}) at 298 K, $\Delta G_f^\ominus$ will be a **large positive** value. [1]

Ammonia can be formed from its elements.



(ii) Suggest whether the ratio of [products]/[reactants] at equilibrium for the formation of ammonia at 298K will be less than, equal to or greater than 1. Give a reason for your answer.

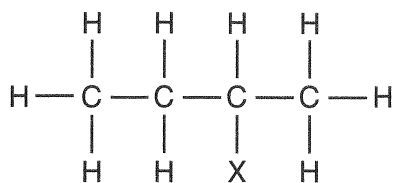
Since $\Delta G_f^\ominus$ is negative, the position of equilibrium lies to the product side (right-hand side), resulting in the ratio of [products]/[reactants] to be **greater than 1**, at 298 K. [1]

[Total: 12]





- 3 Halogenoalkane **B** reacts with sodium hydroxide, under different conditions, to form 4 different organic products **C**, **D**, **E** and **F**.



halogenoalkane **B**

C is the product formed when **B** is warmed with aqueous sodium hydroxide.

- (a) (i) Name the *type of reaction* occurring when **C** is made from **B**.

Nucleophilic substitution.....[1]

- (ii) Describe a chemical test, with appropriate observations, which would confirm the identity of **B** as a chloroalkane.

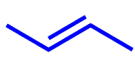
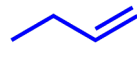
Heat a few drops of B with aqueous sodium hydroxide in a test-tube. Allow the mixture to cool, acidify with dilute nitric acid, and add in aqueous silver nitrate. If B is a chloroalkane, a white precipitate of AgCl is will seen......[3]

- (b) Compounds **D**, **E** and **F** are all hydrocarbons. **D** and **E** are stereoisomers of each other.

- (i) Describe the reagents and conditions needed to favour the formation of a mixture of products **D**, **E** and **F**, rather than product **C**.

Heating B with concentrated ethanolic sodium hydroxide under reflux.....[2]

- (ii) Draw the **skeletal formulae** of **D**, **E** and **F** and name them.

D	E	F
		
Name..... <u>cis-but-2-ene</u>	Name..... <u>trans-but-2-ene</u>	Name..... <u>but-1-ene</u>

[2]





- (iii) Explain how molecules **D** and **E** are stereoisomers of each other but molecule **F** does **not** show stereoisomerism.

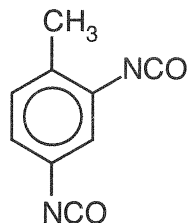
Each carbon of the C=C carries one H atom and one methyl group for **D** and **E**. Despite having identical connectivity, due to restricted rotation about the C=C, **D** with both methyl groups on the same side and **E** with both groups on opposite sides are non-superposable, thus stereoisomeric. **F** on the other hand, has two identical H atoms on one of the C=C carbon, and hence such spatial differentiation of substituents about the C=C is not seen. [2]

[Total: 10]





- 4 (a) 2,4-diisocyanato-1-methylbenzene, 2,4-TDI, is used to make polyurethane foams.



2,4-TDI

It can be made in the 3-step process shown in Fig. 4.1.

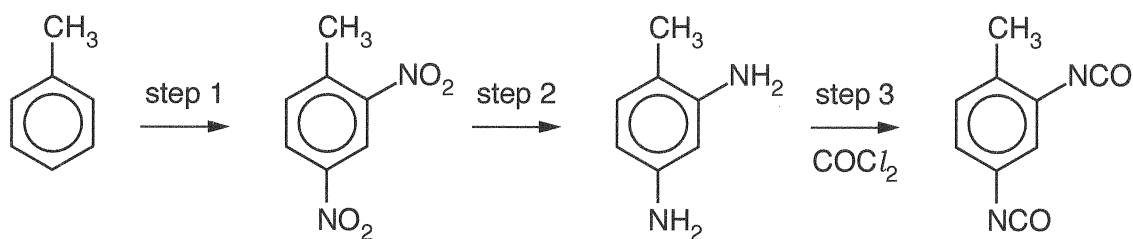


Fig. 4.1

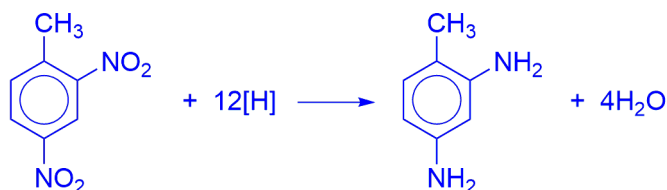
- (i) State the reagents and conditions needed for steps 1 and 2.

step 1 concentrated HNO₃, concentrated H₂SO₄, maintain at 60 °C

step 2 Sn, concentrated HCl, reflux; followed by dilute NaOH

[2]

- (ii) Step 2 is a reduction reaction. Write an equation for step 2 using [H] to represent the reducing agent.



[1]

- (iii) Another organic compound, 2,6-TDI, is often found in the reaction mixture.

2,4-TDI and 2,6-TDI are isomers of each other. Name and define the type of isomerism shown.

Constitutional isomerism since 2,4-TDI and 2,6-TDI have the same molecular formula, but differ in the ways the atoms are bonded together.

or

Positional isomerism due to different relative positioning of the two isocyanate groups with respect to the methyl group on the benzene ring.

[2]

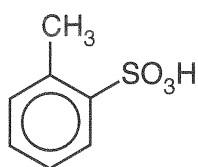
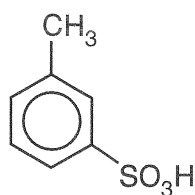
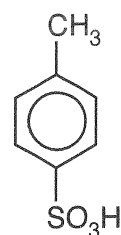
- (iv) Write the molecular formula of 2,6-TDI.

C₉H₆N₂O₂

[1]



- (b) Methylbenzene reacts with fuming sulfuric acid, $\text{H}_2\text{S}_2\text{O}_7$, to form a mixture of three isomers, **G**, **H** and **J**, in equilibrium.

**G****H****J**

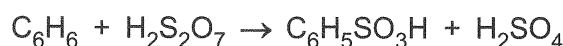
- (i) Explain why **H** is the isomer least likely to be formed in this reaction.

Due to electron-donating effect of the methyl group, which increases electron density at the 2-, 4- and 6-positions, the electron density at C-3/C-5 is the lowest. [1]

- (ii) Suggest why there is a lower concentration of **G** than of **J** in the equilibrium mixture.

Due to steric repulsion between the adjacent CH_3 and SO_3H groups, **G** will be less stable than **J**, hence the equilibrium favours a higher proportion of the more stable **J**. [1]

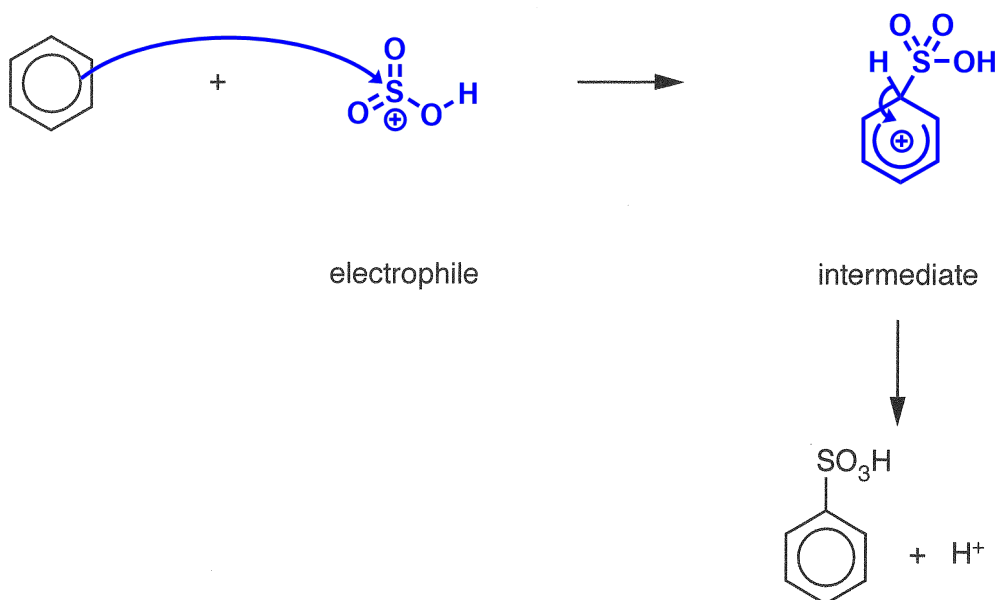
- (c) Under different conditions, benzene will also react with fuming sulfuric acid, $\text{H}_2\text{S}_2\text{O}_7$.



Assume that fuming sulfuric acid ionises as shown.



- (i) Complete Fig. 4.2 to suggest a mechanism for this reaction. Show the displayed structure of the electrophile, the structure of the intermediate and the movement of electron pairs by using curly arrows. [3]

**Fig. 4.2**

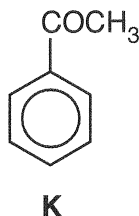


- (ii) When fuming sulfuric acid reacts with alkenes, addition reactions occur.

Explain why benzene does **not** undergo addition reactions with fuming sulfuric acid.

The 6 π electrons in benzene are delocalised throughout the ring, resulting in resonance stabilisation, which is retained when benzene undergoes substitution. On the other hand, addition will result in lost of resonance energy, which is unfavourable. [2]

- (iii) **K** also undergoes a reaction with fuming sulfuric acid.



Suggest the order of reactivity of benzene, methylbenzene and **K** with fuming sulfuric acid. Explain your answer.

Compared to benzene, the CH_3 group of methylbenzene is electron-donating, rendering the benzene ring more electron-rich, while the CH_3CO group of **K** is electron-withdrawing, rendering the benzene ring less electron-rich. Hence the order of increasing reactivity (nucleophilicity) towards the SO_3H^+ electrophile will be **K** < benzene < methylbenzene. [2]

[Total: 15]





Question 5 starts on the next page.





- 5 (a) An *electrolysis cell* looks similar to an electrochemical cell. However, in an electrolysis cell the electrodes are connected to a power supply which drives a chemical reaction.

The Clark Sensor, shown in Fig. 5.1, is an electrolysis cell which is used to measure the concentration of oxygen gas dissolved in samples of water from rivers, seas and blood.

If the external voltage applied to the cell is equal to E_{cell} , then no current flows.

When the external voltage is increased beyond E_{cell} , current will flow and a chemical reaction takes place.

The measurements are taken at 298K.

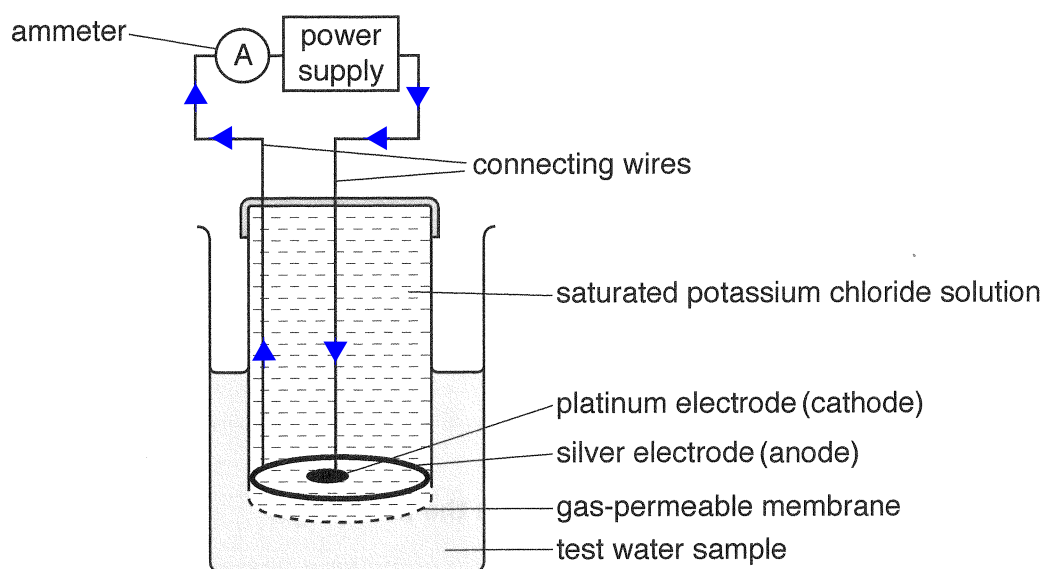
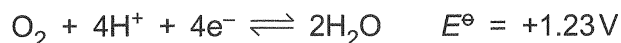
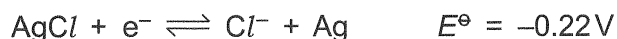


Fig. 5.1

Oxygen gas dissolved in the water sample can diffuse across the gas-permeable membrane.

The two half-equations which are involved in the electrode reactions are



A constant voltage is applied and the rate of flow of electrons produced as these reactions take place is measured by the ammeter. The rate of flow of electrons is proportional to the concentration of oxygen in the water sample.

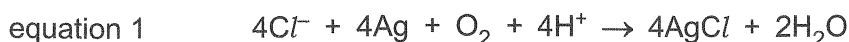
The water sample must be stirred continuously during the measurement.

- (i) Suggest why there is a flow of electrons when the external voltage is +1.45V.

Since concentration of oxygen in the water sample is less than that under standard condition, the E for the reduction process is less than +1.23V, resulting in a smaller E_{cell} than $E_{\text{cell}}^\ominus = +1.45\text{V}$. Hence the external voltage is larger than E_{cell} , giving a current flow [2]



Equation 1 shows the reaction occurring when the measurements are taken.



- (ii) State the type of reaction involving oxygen at the platinum electrode. Describe the change in oxidation state of oxygen which occurs during this reaction.

The oxidation state of oxygen decreases from 0 in O_2 to -2 in H_2O . Hence

reduction is occurring at the platinum electrode. [1]

- (iii) Draw on Fig. 5.1 the direction of electron flow when the reaction shown in equation 1 is occurring. [1]

- (iv) Calculate the rate of reaction of oxygen, in molecules per minute, that produces a current of 0.85A.

$$\begin{aligned}\text{Charge flow per minute} &= I \times t = 0.85A \times 60s \\ &= 51 \text{ C}\end{aligned}$$

$$\begin{aligned}\text{Amt of electrons flow per minute} &= \frac{51 \text{ C}}{96500 \text{ C mol}^{-1}} \\ &= 5.285 \times 10^{-4} \text{ mol}\end{aligned}$$

$$\text{Since } n_{O_2} = \frac{1}{4} n_{e^-}$$

$$\begin{aligned}\text{Rate of reaction of } O_2 &= \frac{1}{4} \times 5.285 \times 10^{-4} \text{ mol min}^{-1} \\ &= \frac{1}{4} \times 5.285 \times 10^{-4} \times 6.02 \times 10^{23} \text{ molecules min}^{-1} \\ &= 7.954 \times 10^{19} \\ &\approx \underline{7.95 \times 10^{19} \text{ molecules min}^{-1}}\end{aligned}$$

[4]

- (v) State, with a reason, the sign of ΔS for the reaction shown in equation 1.

The reaction leads to an overall consumption of 7 mobile aqueous species, hence

the entropy is expected to decrease, that is, ΔS will be negative. [1]

- (vi) Suggest why the water sample is stirred constantly during the measurement.

This is so that the concentration of O_2 at the membrane interface will be homogeneous

with that of the bulk solution, allowing O_2 to diffuse across the membrane. [1]

- (vii) One of the electrodes needs to be cleaned regularly to remove a white solid which forms during the measurement.

Identify the electrode that needs to be cleaned and the white solid formed.

The silver electrode needs to be cleaned as white insoluble silver chloride ($AgCl$) will

coat the electrode as the reaction in equation 1 takes place. [1]





(viii) Suggest

- why the potassium chloride solution is kept saturated.

When the KCl solution is kept saturated, $[Cl^-]$ will be constant at the anode, so that

E_{cell} only varies with $[O_2]$ at the cathode, allowing the $[O_2]$ to be measured.

- the effect on E_{cell} if a much lower concentration of potassium chloride was used. Explain your answer.

With a much lower concentration of KCl, $[Cl^-]$ is much lower than 1 mol dm^{-3} ,

hence E for the anode will be significantly less negative than -0.22V , resulting in a

much less positive E_{cell} compared to $E_{\text{cell}}^\ominus = +1.45\text{V}$.

[3]

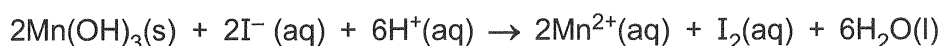
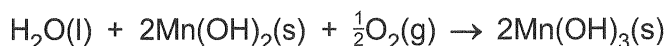
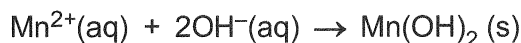
- (b) The **Winkler Method** is another method for determining the oxygen concentration in water samples.

Manganese(II) sulfate is added to an excess of alkali.

The resulting mixture is added to 100 cm^3 of the water sample containing dissolved oxygen.

An excess of acidified potassium iodide is then added.

The following reactions occur.



The mixture is then transferred to a conical flask and titrated with a standard solution of sodium thiosulfate using starch as an indicator.



- (i) State how manganese is showing typical behaviour as a transition element during these reactions.

Mn^{2+} catalyses the oxidation of I^{-} by O_2 , via intermediacy of the +3 oxidation state [1]

- (ii) State the electron configuration of the manganese ion in $\text{Mn}(\text{OH})_3$.

$1s^2$ $2s^2 2p^6 3s^2 3p^6 3d^4$ [1]



- (iii) A 100cm^3 sample of water containing dissolved oxygen required 11.20cm^3 of 0.0100mol dm^{-3} sodium thiosulfate solution to reach the end point. Calculate the concentration of O_2 in this sample in mol dm^{-3} .

$$n_{\text{S}_2\text{O}_3^{2-}} = \frac{11.20}{1000} \times 0.0100 = 1.12 \times 10^{-4} \text{ mol}$$

$$n_{\text{I}_2} = \frac{1}{2} n_{\text{S}_2\text{O}_3^{2-}} = \frac{1}{2} \times 1.12 \times 10^{-4} = 5.60 \times 10^{-5} \text{ mol}$$

$$\text{Since } \frac{1}{2} \text{O}_2 \equiv 2\text{Mn}(\text{OH})_3 \equiv \text{I}_2$$

$$n_{\text{O}_2} = \frac{1}{2} n_{\text{I}_2} = \frac{1}{2} \times 5.60 \times 10^{-5} = 2.80 \times 10^{-5} \text{ mol}$$

$$\begin{aligned} [\text{O}_2] &= \frac{n_{\text{O}_2}}{V} = \frac{2.80 \times 10^{-5} \text{ mol}}{\frac{100}{1000} \text{ dm}^3} \\ &= \underline{2.80 \times 10^{-4} \text{ mol dm}^{-3}} \end{aligned}$$

[3]

Question 5 continues on the next page.





- (c) The concentration of oxygen gas in a 1 dm^3 sample of water saturated with oxygen was measured at different temperatures and the results are shown in Table 5.1.

Table 5.1

sample	temperature / °C	concentration of oxygen at saturation / mg dm^{-3}
1	20	9.07
2	25	8.24
3	30	7.54
4	35	6.93

- (i) The temperature of another 1 dm^3 sample of distilled water saturated with oxygen is raised from 25°C to 35°C and allowed to reach equilibrium.

Calculate the amount of oxygen gas, in moles, which will have escaped from the sample.

$$m_{\text{O}_2} \text{ escaped per dm}^3 = 8.24 - 6.93 = 1.31 \text{ mg}$$

$$n_{\text{O}_2} \text{ escaped from } 1 \text{ dm}^3 \text{ sample} = \frac{1.31 \times 10^{-3} \text{ g}}{16.0 \times 2 \text{ g mol}^{-1}} = 4.09375 \times 10^{-5} \text{ mol}$$

$$\approx \underline{4.09 \times 10^{-5} \text{ mol}}$$

[2]





A sealed 1 dm³ vessel, X, contains pure oxygen gas at 25 °C and 100 kPa.

When the vessel is warmed to 35 °C the new pressure of the oxygen gas is 103.4 kPa.

A sealed 2 dm³ vessel, Y, contains 1 dm³ oxygen gas and 1 dm³ of distilled water saturated with oxygen at 25 °C and a pressure of 100 kPa.

The temperature of vessel Y is increased to 35 °C and a new equilibrium is established.

- (ii) Using this information and your answer to (c)(i), calculate the pressure, in kPa, of oxygen gas in vessel Y at 35 °C. Give your answer to **one** decimal place.

Using ideal gas equation,

$$\begin{aligned} \text{pressure from the oxygen gas escaped, } p &= \frac{nRT}{V} \\ &= \frac{(4.09375 \times 10^{-5}) \times 8.31 \times (35 + 273)}{1 \times 10^{-3}} \\ &= 104.7787 \text{ Pa} \approx 0.1048 \text{ kPa} \end{aligned}$$

$$\begin{aligned} \text{pressure of oxygen gas in vessel Y at 35 °C} &= 103.4 \text{ kPa} + 0.1048 \text{ kPa} \\ &= \underline{\underline{103.5 \text{ kPa}}} \end{aligned}$$

[2]

- (iii) Suggest why the actual value for the pressure recorded in vessel Y was slightly different from the value calculated by this method. Assume oxygen gas behaves as an ideal gas under these conditions.

The oxygen pressure above the water saturated with oxygen changes with temperature.

and hence the solubility of oxygen differs slightly from that in Table 5.1, increasing..... [1]

slightly with higher external pressure.

[Total: 24]





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